MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 20: Friday 23 October 2026

(Week 10)

Remark. Q19.7 is leftover exercise from the Tuesday class.

Remark. Q20.1 to Q20.5 (cylindrical coordinates) are on Section 14.7 of Thomas' Calculus.

Q19.7. Consider the region in the ***xyz***-space given by

$$D = \{(x, y, z) \,:\, 1 \leq x \leq 2,\ 0 \leq xy \leq 2,\ 0 \leq z \leq 1\}.$$

Evaluate

$$\iiint_D (x^2 y + 3xyz)\, dx\, dy\, dz$$

by applying the transformation

$$u = x, \quad v = xy, \quad w = 3z$$

and integrating over an appropriate region in the ***uvw***-space.

Q20.1. Sketch the solid region described by the following inequalities in cylindrical coordinates in the three dimensional space:

$$r \leq z \leq \sqrt{9 - r^2}.$$

Q20.2. Evaluate the following iterated integral in cylindrical coordinates:

$$\int_0^{2\pi} \int_0^{\theta/2\pi} \int_0^{3+24r^2} r \, dz \, dr \, d\theta.$$

Q20.3. Convert the integral

$$\int_{-1}^{1} \int_{0}^{\sqrt{1-y^2}} \int_{0}^{x} (x^2 + y^2)\, dz\, dx\, dy$$

into an integral in cylindrical coordiantes, and then evaluate the resulting integral.

Q20.4. Find the volume of the solid bounded above by the sphere $x^2 + y^2 + z^2 = 9$ and below by the paraboloid $8z = x^2 + y^2$.

[Suggestion: Use cylindrical coordinates.]

Q20.5. A steel beam with a pointed end and a round end is modelled as the set of points in the xyz-space in the cylinder of radius 1 centered at the origin, between the cone $z = x^2 + y^2 - 10$ and the hemisphere $z = \sqrt{1 - x^2 - y^2}$.

From the way that the steel beam is made, you deduce that its density $f(x, y, z)$ at a point (x, y, z) in the steel beam is proportional to the absolute value of its z-coordinate.

You also measured that the total mass of the steel beam is 5.

Find a formula for the density function f.